

Worksheet Ch 1 – Similarity Coefficient (Simple matching, Jaccard etc)

Shoppers' characteristics captured by cameras at a modern IT mall are given in the table below:

	Tall or short	Body size heavy-set or not	Suspected ethnic background	Made purchases or not	Shop alone or not	Fashion conscious or not
Shopper 1	Short	Not	Caucasian	Yes	Alone	Fashionable
Shopper 2	Tall	Heavy	Chinese	No	Alone	Not
Shopper 3	Short	Heavy	Indian	Yes	Alone	Not
Shopper 4	Short	Not	Chinese	No	Alone	Fashionable
Shopper 5	Tall	Heavy	Chinese	No	Not	Not

(a) By introducing appropriate binary variables, calculate **Simple Matching** similarity coefficient for all pairs of shoppers. Present your final answer in a 5 X 5 matrix. (You will need extra writing space to complete your work as this part requires lots of working space).

(more questions on the next page)

(b) Which two shoppers are the most similar based on your answer in part (a)?

(c) Which two shoppers are the least similar?

(d) If you were to divide these 5 shoppers into two relatively homogenous subgroups (clusters) based on the similarity numbers calculated in part (a), how will you form the two subgroups?

(e) Would Simple Matching similarity coefficient or Jaccard similarity coefficient be more suitable for this dataset? Explain.